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Software Engineering

LAB 1: REQUIREMENTS ANALYSIS

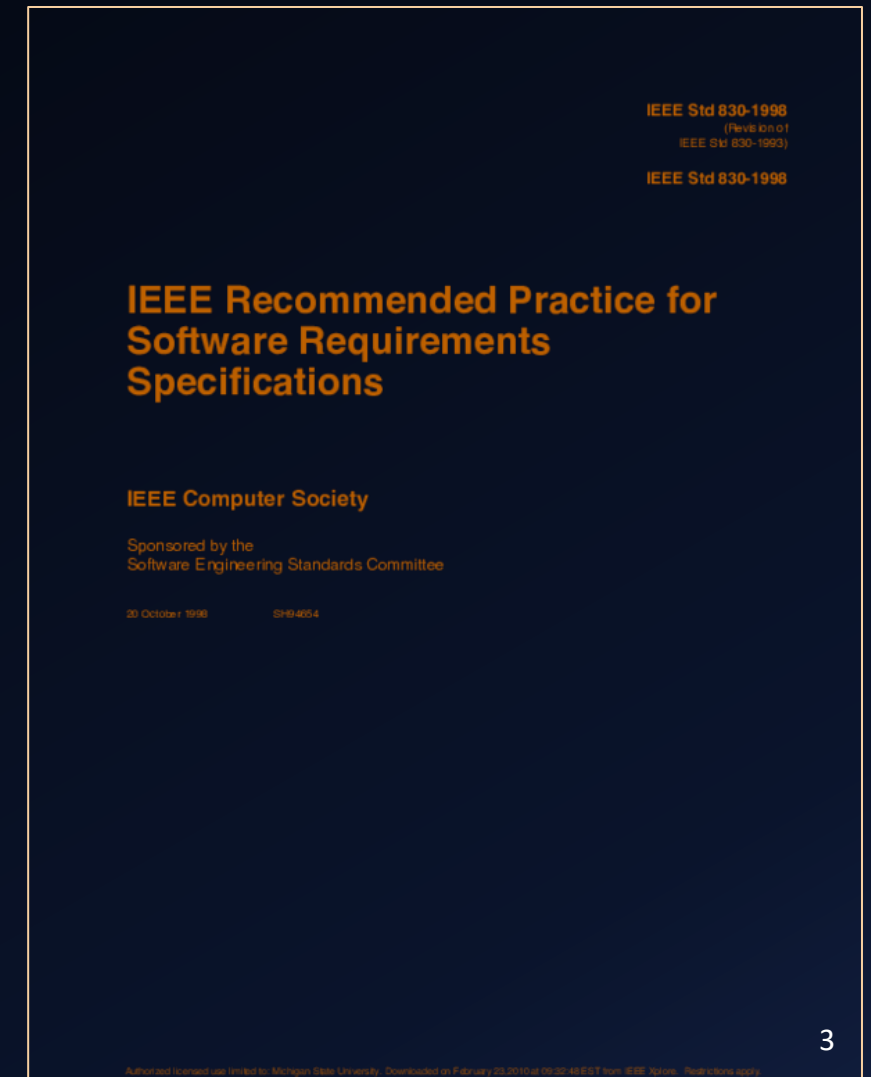
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Lab 1 Overview

- Goal: Transform vague ideas into engineering specifications
- What you will learn:
 - How to write a Software Requirements Specification (SRS)
 - Difference between Functional and Non-Functional Requirements
 - How to make requirements testable and measurable

What is a Software Requirements Specification (SRS)?

- A formal document that describes:
 - **WHAT** the system should do (not HOW)
 - **WHO** will use the system
 - **CONSTRAINTS** and quality attributes
- Based on IEEE 830 standard
- Acts as a contract between developers and stakeholders



The SMART Requirement Criteria

- Every requirement must be:
 - Specific: Describes exactly one behaviour
 - Measurable: Can be tested and verified
 - Attainable: Technically feasible
 - Relevant: Solves a core user problem
 - Traceable: Has a unique ID (e.g., FR-01)

Anatomy of a Good Requirement

- Standard Template:
 - "The [System/Actor] shall [Action] [Object] [Condition]"
- Example:
 - The System shall notify the User via email within 30 seconds of a sensor breach
- Key word: 'shall' = engineering obligation
 - Avoid: 'should', 'can', 'may' (sounds optional)

Bad vs Good Requirements

✗ BAD

- The system should be fast
- The UI must be user-friendly
- The app should work well

✓ GOOD

- The system shall respond in < 2 seconds
- A first-time user shall complete checkout in < 3 minutes
- The system shall maintain 99.5% uptime

Common Pitfalls to Avoid

- Ambiguity: Avoid vague words (fast, robust, efficient)
- Implementation Bias: Don't specify **HOW** yet
 - Bad: 'Use MySQL database to save users'
 - Good: 'The system shall persist user data securely'
- Compound Requirements: Split into testable units
- Untestable Statements: If you can't measure it, it's not valid

Functional Requirements (FR)

- Define WHAT the system does (features and behaviors)
- Examples:
 - FR-01: The system shall allow users to create an account using email
 - FR-02: The system shall display real-time temperature from sensors
 - FR-03: The system shall generate monthly PDF reports
- Minimum: 5 Functional Requirements for your project

Non-Functional Requirements (NFR)

- Define HOW WELL the system performs (quality attributes)
- Categories:
 - Performance: Response time, throughput, concurrent users
 - Security: Encryption, authentication, data privacy
 - Reliability: Uptime %, Mean Time to Recovery
 - Portability: Platform compatibility (Android, iOS, Web)

Non-Functional Requirements - Examples

- NFR-01: The system shall respond to queries in < 2 seconds (Performance)
- NFR-02: All passwords shall be hashed using Argon2 (Security)
- NFR-03: The system shall maintain 99.5% uptime (Reliability)
- Minimum: 3 Non-Functional Requirements for your project

Real-World Example: IoT Sensor System

- Functional Requirement:
 - FR-05: The system shall log sensor data every 10 minutes
- Non-Functional Requirements:
 - NFR-04: The sensor shall operate for 12 months on a single battery
 - NFR-05: Data transmission shall use AES-256 encryption
- This shows requirements aren't just about code—they're about the environment!

SRS Document Structure (IEEE 830)

- 1. Introduction: Purpose, Scope, Audience
- 2. Overall Description: Product perspective, User classes, Constraints
- 3. Functional Requirements: Numbered list (FR-01, FR-02...)
- 4. Non-Functional Requirements: Categorized (NFR-01, NFR-02...)
- 5. External Interfaces: UI, Hardware, Software, APIs

Your Lab 1 Task

- Work in teams (2-3 people) to create an SRS document
- Choose a project: Library System, E-shop, IoT System, etc.
- Write a Problem Statement (0.5 page)
- Define at least 5 Functional Requirements
- Define at least 3 Non-Functional Requirements
- Deliverable: 2-page SRS document (PDF)

Tips for Success

- Use numbers, not adjectives (< 2 seconds, not 'fast')
- Each requirement = ONE action (split if you see 'and')
- Ask: 'How will I prove this works?' = Testability
- Use 'shall' for mandatory, 'may' for optional
- Give each requirement a unique ID for traceability